Training Day 8

1. Topics Covered and Brief Explanation

1. Introduction to Datasets

A dataset is a collection of related data organized in a structured format. In Machine Learning, datasets are used to train, test and evaluate models. Each dataset generally contains rows (records) and columns (features).

2. Types of Data

Studied the different types of data used in AI and Machine Learning:

- Structured Data: Organized in rows and columns, such as spreadsheets and database tables.
- Unstructured Data: Data without a fixed tabular structure, such as images, videos and audio files.
- Semi-Structured Data: Data that does not follow a strict table format but contains organizational elements, such as JSON and XML files.

3. Data Collection Methods

Learned about different methods of collecting data for machine learning applications:

- Surveys and questionnaires.
- Databases and spreadsheets.
- Websites and online repositories.
- Sensors and cameras.
- Publicly available datasets.

4. Features and Labels

- Features: Input variables used by a machine learning model to make predictions.
- Labels: The expected output or target value that a model learns to predict in supervised learning.

For example, in student performance prediction, study hours and attendance can be features, while examination marks can be the label.

5. Importance of Data in Machine Learning

Studied how the quality, quantity and accuracy of data affect the performance of machine learning models. Understood that incorrect or incomplete data can lead to inaccurate predictions.

2. Tools and Technologies Used

Tool	Purpose

Python	Basic data handling and processing.
Pandas	Organizing and exploring datasets.
Google Colab	Working with sample datasets.
Kaggle	Exploring publicly available datasets.
CSV Files	Storing and accessing structured data.

3. Practical Learning / Activities

Explored a sample student performance dataset using Pandas.

Code Example:

```
import pandas as pd
data = {
   "Name": ["Aman", "Riya", "Karan"],
   "Study_Hours": [2, 5, 3],
   "Attendance": [70, 90, 80],
   "Marks": [65, 92, 75]
}
df = pd.DataFrame(data)
print(df)
print("Dataset Shape:", df.shape)
print("Column Names:", df.columns)
```

Explanation:

- Created a sample dataset using a Python dictionary.
- Converted the dictionary into a Pandas DataFrame.
- Identified the features and target label.
- Used shape and columns to understand the dataset structure.

4. Learning Outcomes

- Understood the meaning and importance of datasets in Machine Learning.
- Learned about structured, unstructured and semi-structured data.
- Explored different methods and sources of data collection.
- Understood the difference between features and labels.
- Gained basic experience in creating and exploring datasets using Pandas.